

FleetOps AI

AI Operations Intelligence for Construction Equipment

Updated Project Report

Track: AI + Construction Machinery -> AI + Operations

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1. Executive Summary

Construction fleet operations generate continuous data on operating hours, downtime, fuel use, maintenance, and project utilization. The operational challenge is not simply collecting these records, but converting them into a clear answer to three questions: **what needs attention, why, and what action should be taken next?**

FleetOps AI is a working AI + Operations prototype that combines fleet analytics, machine-learning anomaly detection, a natural-language operations agent, an explainable Attention Score, a 7-day Maintenance Planner, and a What-If Maintenance Simulator. The platform is deployed as a Streamlit web application and can operate in a built-in analytics mode without an external API key, while also supporting an optional OpenAI LLM mode for richer natural-language interaction.

Metric	Current prototype value
Fleet size	30 units
Monitoring period	2026-06-01 - 2026-08-29
Daily records	2,700 records (30 units x 90 days)
Overall utilization rate	95.2%
Total operating hours	15,862.7 hrs
Total downtime	793.0 hrs
Total fuel consumption	293,831.3 L

2. Problem Statement & Objectives

The Problem

Fleet managers often review spreadsheets and periodic reports manually to identify high downtime, abnormal fuel consumption, repeated maintenance, and utilization changes. This is slow, difficult to prioritize, and can delay intervention on equipment that needs attention first.

Project Objectives

- Provide a single operational dashboard for fleet KPIs, downtime, utilization, fuel, and priority signals.
- Allow managers to ask supported fleet questions in natural language and receive data-grounded answers.
- Detect abnormal operating behavior using statistical logic and Isolation Forest machine learning.
- Combine multiple warning signals into an explainable Attention Score (0-100).
- Turn priorities into a practical 7-day maintenance schedule based on the number of units a manager can service.
- Let managers simulate the operational effect of delaying maintenance before making a decision.
- Provide manager-ready reports, recommendations, and downloadable data outputs.

3. Methodology & System Architecture

The latest prototype follows an end-to-end decision-support workflow: **Monitor -> Detect -> Prioritize -> Plan -> Simulate -> Act.**

Layer	Function in the latest version
Data	Daily equipment logs and master data covering equipment, project, hours, downtime, fuel, and maintenance events.
Analytics	Pandas/NumPy KPI calculation for utilization, downtime, fuel efficiency, maintenance frequency, and project performance.
Machine Learning	Isolation Forest detects unusual operational behavior; ML signals are combined with operational indicators.
Priority Engine	Unified Attention Score (0-100) ranks units and explains the strongest reason plus a recommended action.
AI Operations Agent	Built-in intent routing and analytics tools; optional OpenAI Function Calling uses the same underlying tools.
Decision Support	7-day Maintenance Planner and What-If Maintenance Simulator convert analytics into operational choices.
Interface & Output	Streamlit dashboard, structured answer cards, charts, report downloads, and user data upload.

Core Analysis Tools

- **get_top_downtime_equipment(top_n)** - returns the requested number of highest-downtime units.
- **get_equipment_needing_attention(top_n)** - returns priority units with a reason and recommended action.
- **explain_utilization_trend(project)** - analyzes the whole fleet or a specifically named project.
- **analyze_fuel_consumption(equipment_id)** - detects abnormal fuel-consumption changes and validates equipment IDs.
- **compute_attention_scores(top_n)** - ranks units using the unified Attention Score with explanations.
- **build_maintenance_plan(top_n, horizon_days)** - builds a 7-day maintenance schedule from the highest-priority units.
- **simulate_maintenance_delay(equipment_id, delay_days)** - estimates the operational impact of delaying maintenance.
- **generate_fleet_report()** - generates a fleet summary with exactly three manager actions.

Technology Stack

Layer	Tools
Core development	Python
Data analysis	Pandas, NumPy
Machine learning	scikit-learn (Isolation Forest)
AI interaction	Built-in routing + optional OpenAI Function Calling
Visualization	Plotly, Matplotlib
Web interface	Streamlit
Deployment	GitHub + Streamlit Community Cloud

4. Data

The prototype uses a synthetic construction-fleet dataset created for demonstration and controlled validation. It represents **30 machines, 6 equipment types, 5 active projects, and 90 days of daily logs**, resulting in 2,700 operational records. The dataset intentionally includes known issues so that the system's detection and prioritization logic can be checked against expected behavior.

- Three units with unusually high downtime.
- One unit (EQ-008) with a deliberately injected fuel-consumption increase.
- One project with a utilization decline pattern.
- Repeated maintenance events on selected units for priority-testing scenarios.

The schema mirrors common telematics/ERP exports, so the application also includes an **Upload Your Data** workflow for compatible real datasets.

5. Results & Operational Insights

The latest system identifies operational issues and links them to recommended actions rather than returning raw metrics only.

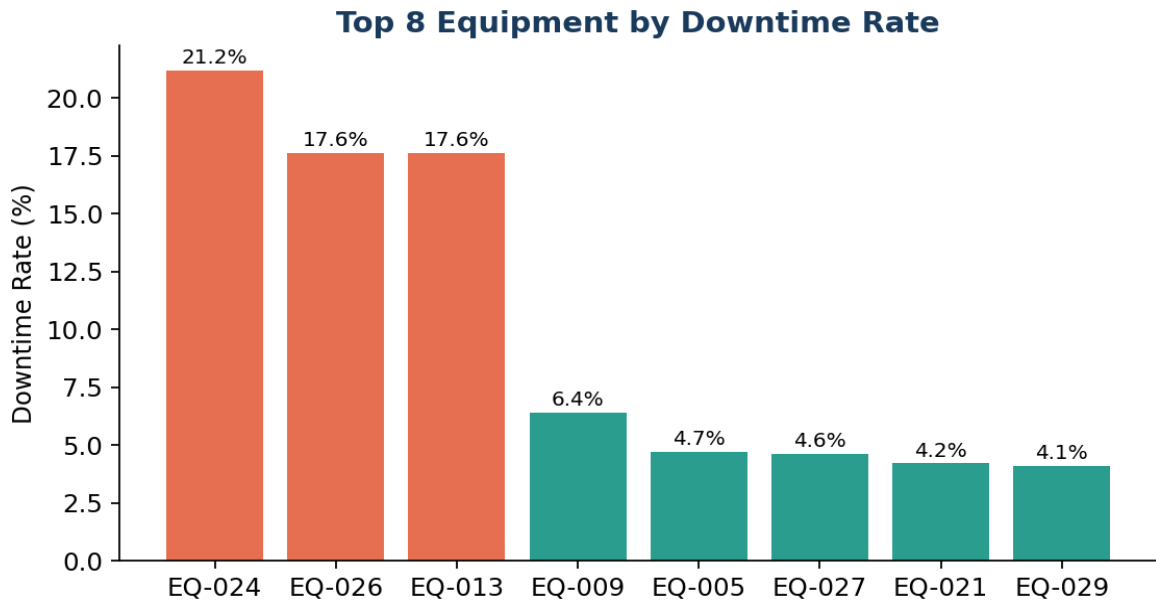


Figure 1 - Equipment ranked by downtime. EQ-024, EQ-026, and EQ-013 are the three highest-downtime units.

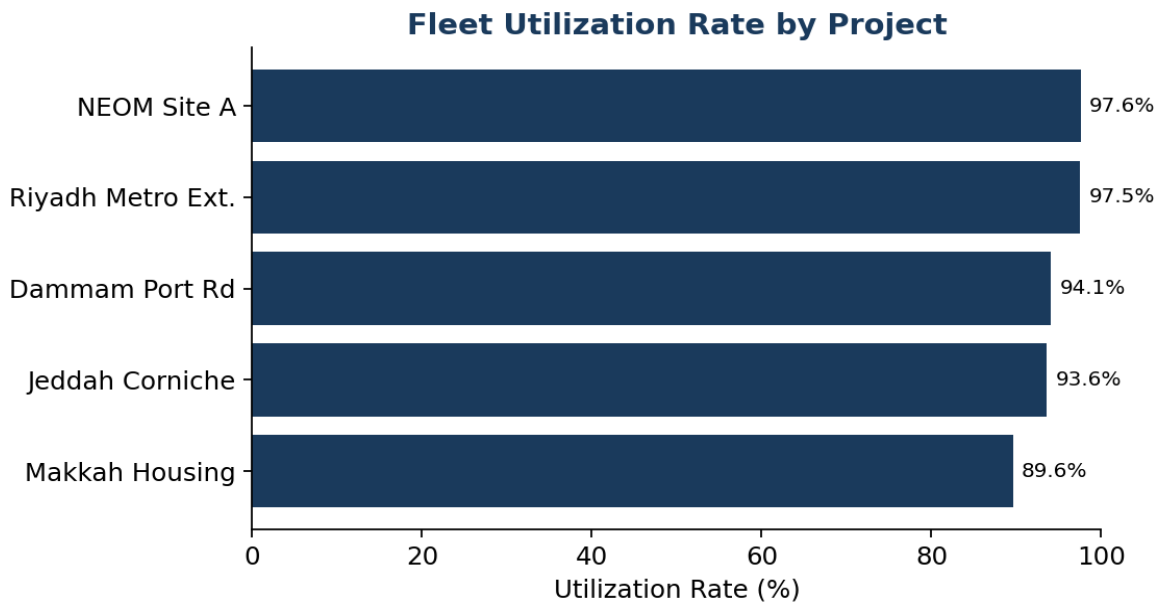


Figure 2 - Utilization by project. Makkah Housing has the lowest current project utilization at 89.6%.

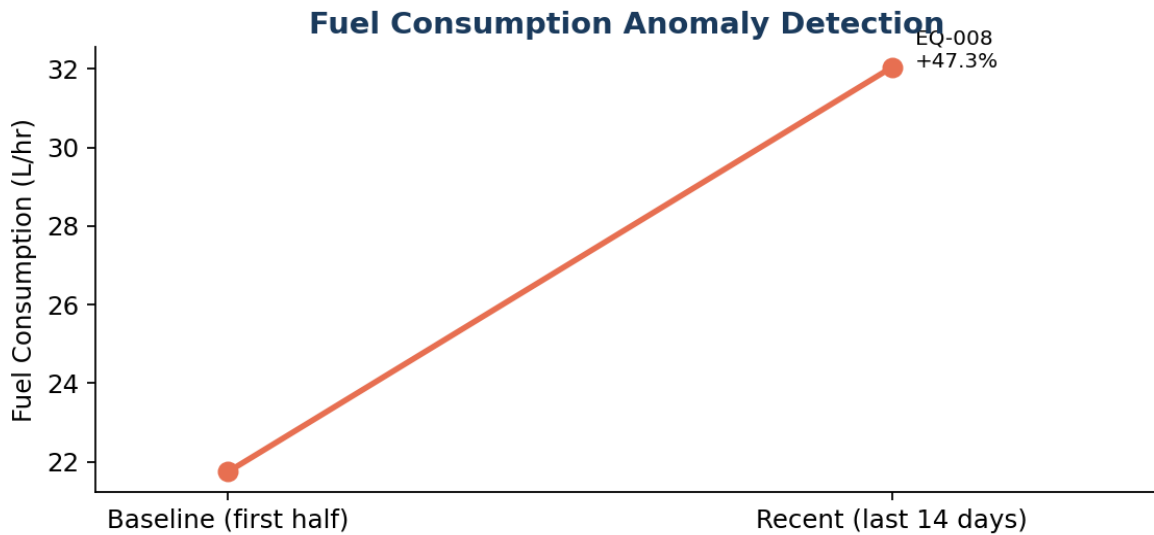


Figure 3 - EQ-008 shows a 47.3% abnormal increase in fuel consumption versus its baseline.

Current Top Operational Issues

Unit	Type	Project	Key issue	Recommended action
EQ-024	Dump Truck	Dammam Port Rd	High downtime (21.2%)	perform an immediate diagnostic inspection and schedule corrective maintenance to reduce downtime
EQ-026	Dump Truck	Jeddah Corniche	High downtime (17.6%)	perform an immediate diagnostic inspection and schedule corrective maintenance to reduce downtime
EQ-013	Grader	Makkah Housing	High downtime (17.6%)	perform an immediate diagnostic inspection and schedule corrective maintenance to reduce downtime

6. Decision-Support Features Added in the Latest Version

A. AI Maintenance Planner

The manager selects how many equipment units can be included in the 7-day maintenance plan. The system then automatically selects the highest-priority units using the Attention Score and assigns earlier days to higher-risk equipment. Estimated service hours are derived from maintenance-day downtime observed in the dataset.

Day	Priority	Unit	Project	Attention	Est. service
Day 1	Critical	EQ-026	Jeddah Corniche	56.1/100	3.7 h
Day 1	Critical	EQ-024	Dammam Port Rd	53.6/100	4.5 h
Day 1	Critical	EQ-008	Riyadh Metro Ext.	52.6/100	3.8 h
Day 2	High	EQ-013	Makkah Housing	45.7/100	4.8 h
Day 3	High	EQ-004	NEOM Site A	35.8/100	3.7 h

Example (Top 5): 3 Critical, 2 High Priority, 5 scheduled units, with approximately 20.5 service hours.

B. What-If Maintenance Simulator

The simulator lets a manager choose one unit and test a maintenance-delay scenario. It is a transparent scenario estimate - not a failure-probability prediction. The model stress-tests observed daily downtime using the current Attention Score and selected delay length, then shows the change in expected downtime and utilization.

Scenario	Value
Equipment	EQ-024 - Dump Truck
Delay	9 days
Attention Score	53.6/100 (High impact)
Expected downtime	11.25 h -> 13.19 h
Additional downtime	+1.94 h
Utilization	78.8% -> 76.0% (-2.8 pp)
Recommended decision	Prioritize maintenance now; delaying increases the operational downtime exposure.

C. Structured AI Answer Cards

Important AI-agent responses are displayed as structured cards instead of long text blocks. For priority questions, each card shows the equipment ID, type, key metric, reason, and next action. The full analytical explanation remains available in an expandable section.

7. AI Agent, Reporting & User Experience

- Built-in AI analytics mode works without an API key for supported operational intents.
- Optional OpenAI mode can use Function Calling while keeping the same analytical tools underneath.
- Explicit Top-N requests are respected for downtime, maintenance attention, and Attention Score ranking.
- Project-specific utilization questions preserve the named project instead of falling back to the whole fleet.
- Invalid equipment IDs return a clear equipment-not-found response instead of unrelated results.
- Arabic maintenance-priority questions are routed to actionable Arabic answers.
- The Report tab produces fleet summaries, exactly three manager actions, and downloadable CSV/JSON/PDF outputs.

Current Fleet Report - Three Manager Actions

1. Prioritize diagnostic inspection and corrective maintenance for EQ-024 (21.2% downtime), the highest downtime unit.
2. Schedule follow-up diagnostics for the next high-downtime units: EQ-026, EQ-013.
3. Inspect the fuel system of EQ-008 after a 47.3% abnormal consumption change.

8. Validation & Testing

The latest package includes a functional regression suite covering data integrity, KPI accuracy, fuel anomaly detection, ML cross-validation, agent routing, edge cases, Top-N behavior, project-specific filtering, invalid equipment IDs, Arabic responses, the Maintenance Planner, the What-If Simulator, and structured card responses.

The core application and decision-support features pass their functional checks. The synthetic dataset also contains known ground-truth anomalies, allowing the team to verify that the system detects the intended operational patterns.

Validated capability	Observed result
Fuel anomaly	EQ-008 correctly detected after a 47.3% consumption increase
Downtime ranking	Top-3 request returns exactly EQ-024, EQ-026, EQ-013
Project utilization	Makkah Housing is analyzed as its own project scope
Invalid equipment	Unknown IDs return equipment_not_found
Maintenance Planner	7-day prioritized schedule with actions and service-hour estimates
What-If Simulator	Delay scenario returns impact band, downtime change, utilization change, and recommendation
Structured AI response	Priority results render as cards with reasons and next actions

9. Current Project Status & Future Development

Component	Status
Synthetic fleet dataset + KPIs	Complete
Dashboard + charts	Complete
Built-in AI operations agent	Complete
Optional LLM integration	Complete
Isolation Forest + Attention Score	Complete
Structured AI answer cards	Complete
7-day Maintenance Planner	Complete
What-If Maintenance Simulator	Complete
Report exports + data upload	Complete
Cloud deployment	Complete

Future Development

- Connect directly to live telematics/ERP feeds and maintenance work-order systems.
- Add predictive failure models once sufficient labeled historical failure data is available.
- Add maintenance-capacity constraints (technicians, bays, shifts, spare parts) to optimize daily schedules.
- Expand natural-language intent coverage to additional fleet questions and cross-project comparisons; a known example is the currently unsupported question about the most-used equipment in each project, which can fall back to a generic fleet report in built-in mode.
- Add proactive notifications when an Attention Score or anomaly crosses an operational threshold.

10. Conclusion

FleetOps AI has progressed from a monitoring dashboard into a practical AI + Operations decision-support system. The latest version does not stop at identifying a problem: it prioritizes the equipment, explains the reason, recommends an action, schedules maintenance, and allows the manager to test the impact of delaying that maintenance. This creates a clear path from raw fleet data to an operational decision a manager can act on today.

The current prototype is intentionally transparent about its synthetic dataset and scenario estimates, while its modular architecture is ready to accept compatible real operational data and additional AI/optimization capabilities in future deployment.